

California Statewide Fires Situation Report*

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*This report was prepared as part of the partnership between Carnegie Mellon University and Meta's AI for Good.

Disaster Manager Brief - California Statewide Fires

Overview

Where Did Population Changes Occur?

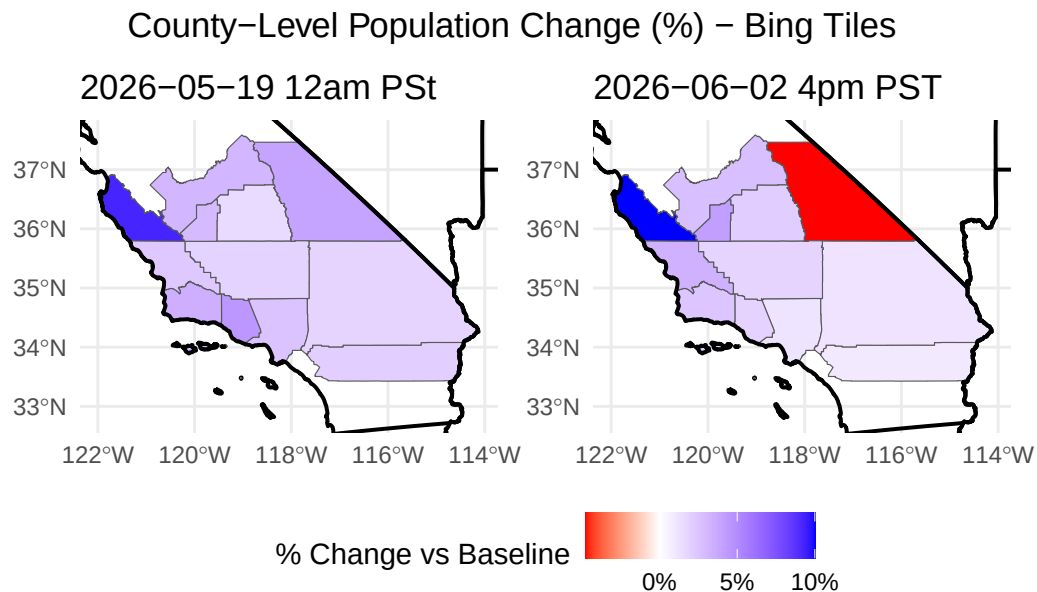


Figure 1: Red shades represent counties where the population decreased relative to the baseline number (i.e., a population exodus), whereas blue shades represent population increases relative to baseline number (i.e., population influx). Population changes are measured relative to a baseline population level computed from the 45 days preceding the event...

...

Takeaways (Figure 1)

- TBD ¹

¹Baseline count refers to the average number of Facebook Location Services-enabled users observed during the pre-event reference period. Values shown here have been rounded to the nearest integer. Additional details are provided in the Facebook Data Definitions section.

Counties Experiencing the Largest Changes

% Population Change Over Time: Highest-Change Counties

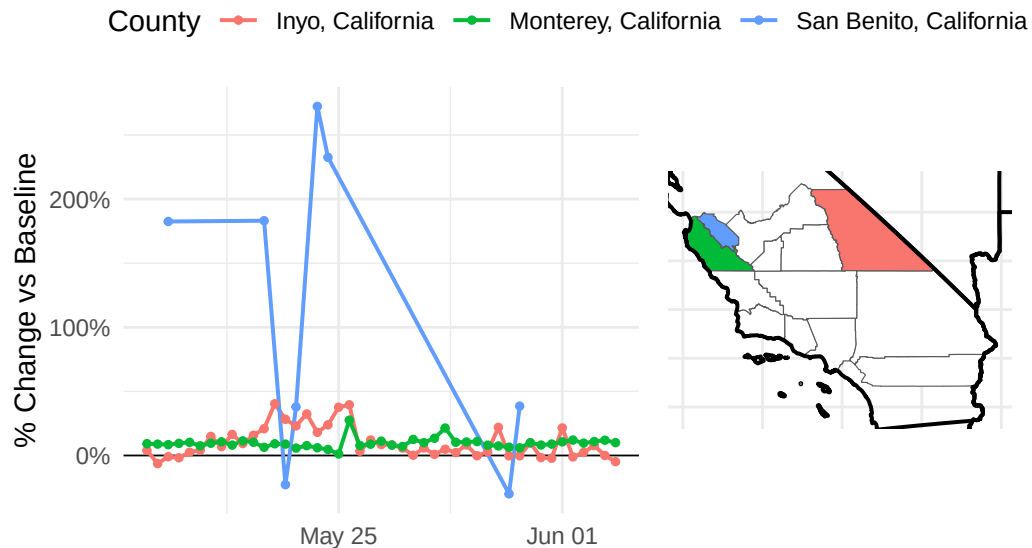


Figure 2: The figure shows that “highest change” includes both large increases and large decreases when counties are selected using largest absolute percent change in population. The time period is May 19nd to June 2nd at 8hr intervals at 12am PST, 8am PST, and 4pm PST. Percent population change is relative to the baseline population (i.e., 45 days prior to the start of a disaster)

TBD...

Takeaways (Figure 2)

- TBD...

Community Characteristics of Affected Counties

TBD...

1. TBD...
2. (1)
3. (2)

4. (3)

Comprehensive Situation Report

Formatted based on LA Fire [Situation Report](#)

Summary

TBD...

The following visualizations are developed based on Meta AI for Good's data through Facebook. The data covers population movement during the collection period of May 19th to June 2nd. These datasets use location activity from Facebook users to estimate how populations shift during a crisis. Data do not represent the entire population in the area. Counts of people within a certain region are recorded at 8-hour intervals in this dataset. The counts during the crisis are compared to the counts at a baseline, which are the counts in the same region 45 days prior to data collection. These regions are either Bing tiles, what are about 1.5 x 1.5 mile blocks, or administrative regions as determined by [GADM](#). These counts are aggregated to the county level for this analysis.

TBD...

How the Data Are Constructed

The Facebook data used in this report comes from Meta's Data for Good crisis datasets. In plain terms, these datasets use location activity from Facebook users to estimate how population shifts during a crisis. The data does not represent everyone in the affected area: for population and movement datasets, it only represents Facebook app users who have Location Services enabled.

The data is constructed by comparing what is observed during the crisis period to what was typical before the crisis. Namely, population movements during the storm is compared to a pre-crisis baseline of 45-days prior to data collection. Because the data is relative to a baseline, the main signal is not the raw count itself, but whether a place is above or below its pre-crisis level.

Temporally, the population and movement datasets use fixed 8-hour windows. These windows begin at 00:00, 08:00, and 16:00 Pacific Time (PST). This means that the time periods do not automatically adjust to local time zones. For our analysis, we report time windows in Eastern Time (EST).

Some Meta crisis datasets are provided directly as Bing tile data, while others are provided as administrative-region data. **For the rest of this report, we use the Bing tile version as the primary aggregation source because Bing tiles are strictly defined grid**

cells with clear latitude and longitude coordinates, and they are more granular than administrative regions. This makes the spatial assignment process more transparent and reproducible. In contrast, the administrative-region datasets are aggregated using boundaries provided to Meta by [GADM](#). Because we do not have direct access to the exact GADM polygons used by Meta for this aggregation, it is harder to verify or reproduce the administrative-region boundaries precisely.

When the data is provided as Bing tiles, each observation is tied to a grid cell with a latitude and longitude, which is defined as the polygon centroid point. We assign each Bing centroid point to the county that contains it. This allows the tile-level data to be summarized at the county level while preserving a clear record of how the geographic aggregation was performed.

For example, a Bing tile located inside Davidson County, Tennessee would be assigned the Davidson County name and GEOID. If many Bing tiles fall within Davidson County, their values can then be averaged, summed, or otherwise summarized to produce a county-level measure. This approach gives us a consistent way to move from tile-level observations to county-level reporting while relying on publicly available county boundary data. Note that all Bing tiles are smaller than counties in the data sets and hence all counties are accounted for in the merge.

Facebook Data Definitions

- **Baseline Population (n_baseline):** Average number of Facebook Location Services-enabled users present in a Bing tile during the baseline period prior to the event. Baseline values are calculated separately for each day of the week and 8-hour time window using historical observations before the start of data collection. Values below Meta’s privacy threshold are suppressed. Missing values are later imputed as part of the county-level analysis using the imputation procedure described in Section .
- **Crisis Population (n_crisis):** Number of Facebook Location Services-enabled users present in a Bing tile during a given 8-hour reporting window. To protect privacy, the counts are subject to noise addition, spatial smoothing, and suppression of small counts. Observations with counts below Meta’s reporting threshold may appear as missing values. Missing values are later imputed as part of the county-level analysis using the imputation procedure described in Section .
- **Population Change (%):** Percent change in Facebook Location Services-enabled user population counts per tile, calculated by dividing the difference by the baseline number (plus a small value, usually 1)
- **Time Windows:** aggregated over 3 fixed 8-hour time windows, beginning at 00:00, 08:00 and 16:00 PST, that represent changes in population

- **Bing Tiles:** Bing tiles divide the map into a grid of smaller spatial units. Each tile is identified by tile coordinates and a quadkey, which uniquely identifies a tile at a specific level of detail (**microsoft_bing_tile_system?**). In the Meta crisis datasets used here, Bing tile level 14 is roughly 1.5 miles on each side near the equator.
- **Administrative Regions:** the polygon boundaries for administrative regions derived from the [Database of Global Administrative Areas](#), also known as GADM (**gadm?**). Note that we do not have access to these polygon shapes; hence we only use Bing tiles in this report.
- **Percent change:** Percent change in population counts of Facebook app users who have turned on the Location Services device setting on their mobile device and are along the vector defined by the starting Bing tile and ending Bing tile, calculated as the polygon movement at a given time window minus the baseline number (i.e., count 45 days prior to data collection period), divided by the baseline number.

Aggregation Choices

Bing tile observations are assigned to counties using the latitude and longitude coordinates of each tile centroid. The documentation defines these coordinates as the centers of the Bing tile grid cells. These centroid points are spatially joined to county polygons to assign county names and GEOIDs. Quadkeys identify the original Bing tiles, but they are not used directly for county assignment in this report.

Imputation and County-Level Percent Change

For the county-level analysis, this report does not use the original tile-level **percent_change** field as the final county-level measure. Instead, the original **percent_change** column is preserved as **given_percent_change**, and a new county-level **percent_change** is calculated from the underlying baseline and crisis counts.

Some rows have missing values for **n_baseline** or **n_crisis** because of Meta’s privacy protections around small counts (counts less than or equal to 10 are not reported) . To avoid dropping these rows entirely, missing **n_baseline** values are imputed as 3 and missing **n_crisis** values are imputed as 9. See Table 5 for percentage of data imputed.

After imputation, observations are summarized by county and reporting window. Because the data may include multiple observations within a single day, the reporting window is defined using both the date and the time-window information available in the data. For each county-window pair, the baseline counts are summed across all Bing tile observations in that county and reporting window, and the crisis counts are summed across those same observations. The county-window percent change is then calculated as:

$$\text{Percent Change}_{cw} = \frac{\sum_{i \in \text{Bing}_c} n_{\text{crisis},ciw} - \sum_{i \in \text{Bing}_c} n_{\text{baseline},ci}}{\sum_{i \in \text{Bing}_c} n_{\text{baseline},ci} + 1} \times 100$$

where c is the county, w is the reporting window and Bing_c denotes the set of Bing tiles assigned to county c (based on centroid location). The small value added to the denominator follows the logic in the Meta documentation and prevents division by very small baseline values.

This means the maps and tables represent county-level population change based on aggregated counts.

The aggregation method depends on the type of variable being summarized. For count variables, such as `n_baseline` and `n_crisis`, summing is appropriate because the goal is to estimate the total observed Facebook users across tiles within a county. For relative variables, such as `given_percent_change`, or ratios, averaging can still be useful for descriptive comparisons, but these are not used as the main county-level percent change measure in this report.

For movement data, origin and destination counties are kept separate. Each row represents movement from one starting location to one ending location, so the analysis preserves both `start_county_geoid` and `end_county_geoid`. This avoids collapsing directional movement into a single county and allows county-to-county movement patterns to be analyzed more accurately.

Outside Data Added

In addition to the Meta crisis datasets, county-level demographic data was added from the American Community Survey in 2022 ([\(\(acs?\)\)](#)). This demographic data was incorporated with the use of the `tidycensus` package ([\(\(tidycensus?\)\)](#)). This outside data provides context about the counties affected by the storm, including total population, median household income, poverty rate, share of residents age 65 and older, and share of households without vehicle access. These variables help interpret which communities may be more vulnerable during a crisis.

The demographic data is joined using GEOID, a standardized county identifier. This is important because county names alone are not unique across states. For example, many states have counties with the same name, but each county has a unique GEOID.

Limitations

These datasets should be interpreted as indicators of crisis-related change, not exact measures of the full population. The population and movement data only include Facebook users with

Location Services enabled, so the sample may not represent all residents equally. Areas with fewer Facebook users or lower Location Services usage may be less reliable.

Privacy protections also affect interpretation. Small counts at the Bing tile level (less than 11 users) are removed, which can especially affect rural areas. For movement data, this means it is not reliable to calculate exact net flows by simply adding inbound and outbound movement, because some small movement vectors are excluded. To mitigate the effects of these missing values, this report employs a simple imputation procedure. The resulting imputations affect only about 0.6% of Bing tile observations and account for approximately 0.1–0.4% of total population counts in the selected reporting windows (see Table 5).

The baseline period is limited to the pre-crisis period available to Meta. This means the analysis does not have a long historical record of pre-crisis behavior. As a result, seasonal changes, holidays, or unusual pre-crisis conditions may affect the baseline. The data is useful for rapid situational awareness, but it should not be interpreted as a complete long-term population or economic dataset.

Finally, the data is time-limited. The documentation notes that some datasets are accessible for 90 days after the final update day and then discarded, which limits longer-term reproducibility and longitudinal analysis.

Movement

% Population Change

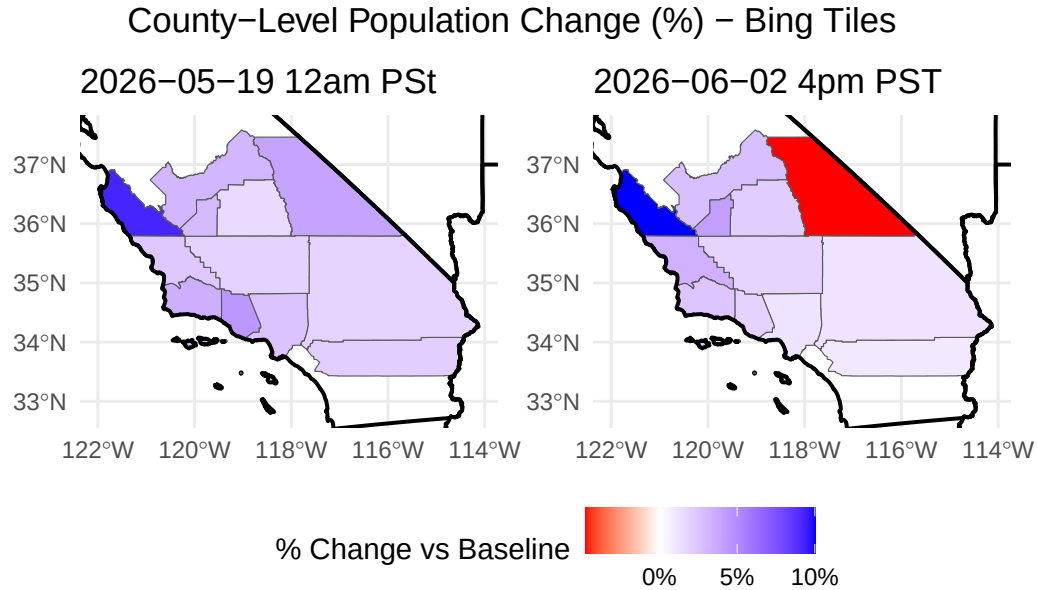


Figure 3: Red shades represent counties where the population decreased relative to the baseline number (i.e., a population exodus), whereas blue shades represent population increases relative to baseline number (i.e., population influx). Population changes are measured relative to a baseline population level computed from the 45 days preceding the event. ...

Takeaways (Figure 3)

- TBD... (1):
- TBD... (2):
- TBD... (Table 3):
- TBD... (Table 4):

Time Series of Movement

The counties are grouped by percent population change over the recording period.²

% Population Change Over Time: Most Extreme-Change Counties

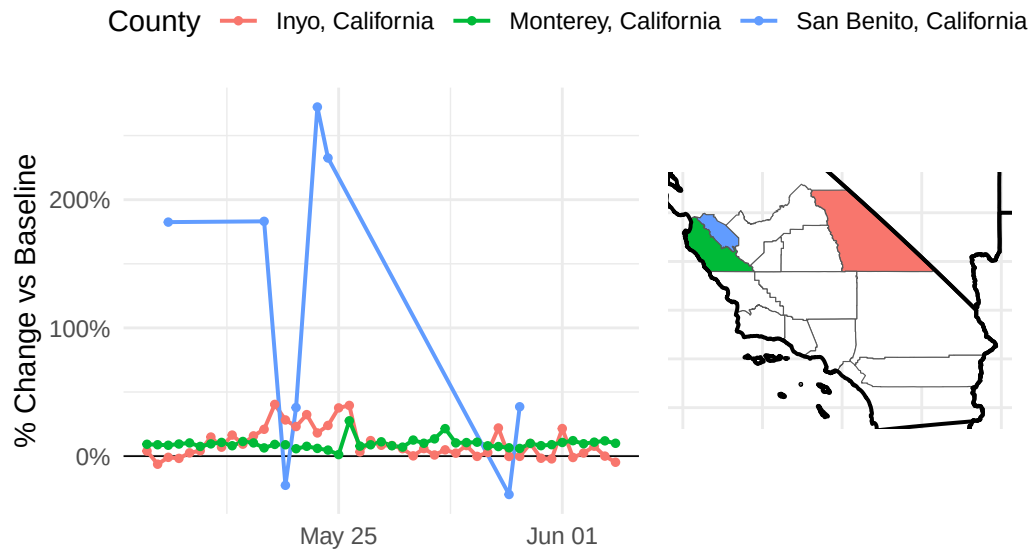


Figure 4: The figure shows that “highest change” includes both large increases and large decreases when counties are selected using largest absolute percent change in population. The time period is May 19th to June 2nd at 8hr intervals at 12am PST, 8am PST, and 4pm PST.

Takeaways (Figure 4)

- TBD...

²See “Percent Change” in Facebook Data Definitions

% Population Change Over Time: Average-Change Counties

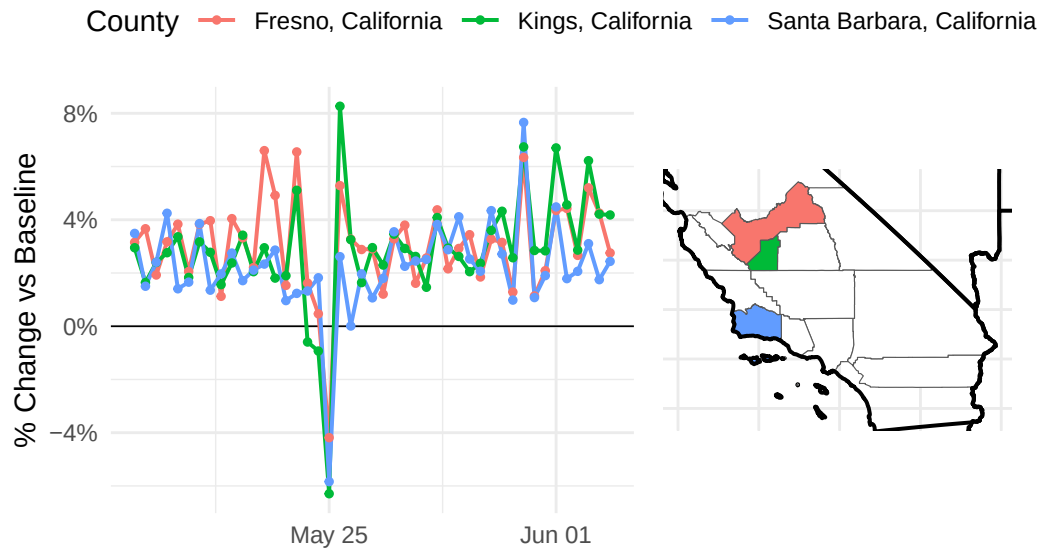


Figure 5: The figure shows that “average change” includes both large increases and large decreases when counties are selected using median absolute percent change in population. ...

Takeaways (Figure 5)

- TBD...

% Population Change Over Time: Lowest-Change Counties

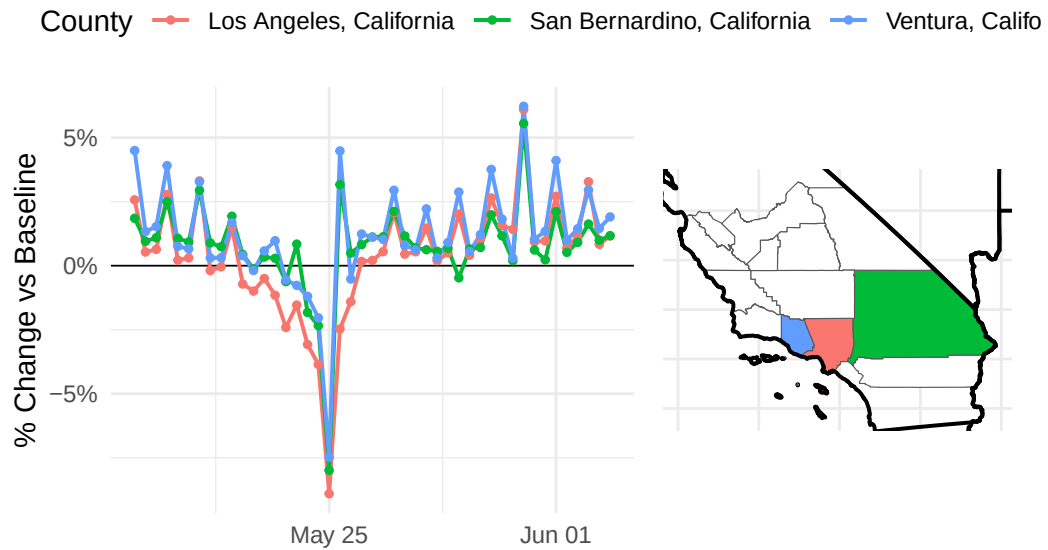


Figure 6: The figure shows that “lowest change” includes one large decrease when counties are selected using lowest absolute percent change in population....

Takeaways (Figure 6)

- TBD...

Counties with Most Increased Population on 2026-05-19 12am PST

Source: Bing Tiles

County	State	Population Change (%) ↓	Total Population	Median Household Income	Poverty Rate (%)	% Age 65+	% No Vehicle Access
Monterey	California	9.2%	437,613	\$97,230	12.7%	1.9%	4.7%
NA	NA	5.8%	NA	NA	NA	NA	NA
Ventura	California	4.5%	837,469	\$109,797	9.3%	2.5%	4.1%
Inyo	California	3.9%	18,739	\$73,991	10.9%	3.4%	5.5%
Santa Barbara	California	3.5%	443,701	\$98,161	14.7%	2.2%	5.9%
Fresno	California	3.2%	1,016,725	\$74,201	18.3%	1.8%	7.0%
Kings	California	2.9%	153,298	\$70,995	16.0%	1.6%	5.8%
Los Angeles	California	2.6%	9,808,667	\$90,112	13.7%	2.2%	8.9%
San Luis Obispo	California	2.4%	281,555	\$97,446	12.9%	2.8%	3.8%
Riverside	California	2.1%	2,478,600	\$93,074	10.6%	2.1%	4.3%
Kern	California	1.9%	915,075	\$70,210	19.1%	1.8%	6.7%
San Bernardino	California	1.9%	2,197,104	\$85,478	13.3%	1.9%	4.6%
Tulare	California	1.6%	478,693	\$71,300	17.9%	1.8%	4.5%

Table 1: ... All demographic information comes from the American Community Survey in 2022. County-level population change is calculated from aggregated baseline and crisis counts across all Bing tiles assigned to each county and reporting window.

Demographic Profile

Counties with Most Decreased Population on 2026-05-19 12am PST

Source: Bing Tiles

County	State	Population Change (%) ↑	Total Population	Median Household Income	Poverty Rate (%)	% Age 65+	% No Vehicle Access
Tulare	California	1.6%	478,693	\$71,300	17.9%	1.8%	4.5%
San Bernardino	California	1.9%	2,197,104	\$85,478	13.3%	1.9%	4.6%
Kern	California	1.9%	915,075	\$70,210	19.1%	1.8%	6.7%
Riverside	California	2.1%	2,478,600	\$93,074	10.6%	2.1%	4.3%
San Luis Obispo	California	2.4%	281,555	\$97,446	12.9%	2.8%	3.8%
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Kings	California	2.9%	153,298	\$70,995	16.0%	1.6%	5.8%
Fresno	California	3.2%	1,016,725	\$74,201	18.3%	1.8%	7.0%
Santa Barbara	California	3.5%	443,701	\$98,161	14.7%	2.2%	5.9%
Inyo	California	3.9%	18,739	\$73,991	10.9%	3.4%	5.5%
Ventura	California	4.5%	837,469	\$109,797	9.3%	2.5%	4.1%
NA	NA	5.8%	NA	NA	NA	NA	NA
Monterey	California	9.2%	437,613	\$97,230	12.7%	1.9%	4.7%

Table 2: ... All demographic information comes from the American Community Survey in 2022. County-level population change is calculated from aggregated baseline and crisis counts across all Bing tiles assigned to each county and reporting window.

Counties with Most Increased Population on 2026-06-02 4pm PST

Source: Bing Tiles

County	State	Population Change (%) ↓	Total Population	Median Household Income	Poverty Rate (%)	% Age 65+	% No Vehicle Access
Monterey	California	10.0%	437,613	\$97,230	12.7%	1.9%	4.7%
NA	NA	4.4%	NA	NA	NA	NA	NA
Kings	California	4.2%	153,298	\$70,995	16.0%	1.6%	5.8%
San Luis Obispo	California	3.3%	281,555	\$97,446	12.9%	2.8%	3.8%
Fresno	California	2.8%	1,016,725	\$74,201	18.3%	1.8%	7.0%
Santa Barbara	California	2.4%	443,701	\$98,161	14.7%	2.2%	5.9%
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Ventura	California	1.9%	837,469	\$109,797	9.3%	2.5%	4.1%
Kern	California	1.8%	915,075	\$70,210	19.1%	1.8%	6.7%
San Bernardino	California	1.2%	2,197,104	\$85,478	13.3%	1.9%	4.6%
Los Angeles	California	1.2%	9,808,667	\$90,112	13.7%	2.2%	8.9%
Riverside	California	0.9%	2,478,600	\$93,074	10.6%	2.1%	4.3%
Inyo	California	-4.8%	18,739	\$73,991	10.9%	3.4%	5.5%

Table 3: ... All demographic information comes from the American Community Survey in 2022. County-level population change is calculated from aggregated baseline and crisis counts across all Bing tiles assigned to each county and reporting window.

Counties with Most Decreased Population on 2026-06-02 4pm EST

Source: Bing Tiles

County	State	Population Change (%) ↑	Total Population	Median Household Income	Poverty Rate (%)	% Age 65+	% No Vehicle Access
Inyo	California	-4.8%	18,739	\$73,991	10.9%	3.4%	5.5%
Riverside	California	0.9%	2,478,600	\$93,074	10.6%	2.1%	4.3%
Los Angeles	California	1.2%	9,808,667	\$90,112	13.7%	2.2%	8.9%
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Kings	California	4.2%	153,298	\$70,995	16.0%	1.6%	5.8%
NA	NA	4.4%	NA	NA	NA	NA	NA
Monterey	California	10.0%	437,613	\$97,230	12.7%	1.9%	4.7%

Table 4: ... All demographic information comes from the American Community Survey in 2022. County-level population change is calculated from aggregated baseline and crisis counts across all Bing tiles assigned to each county and reporting window.

Demographic Correlations

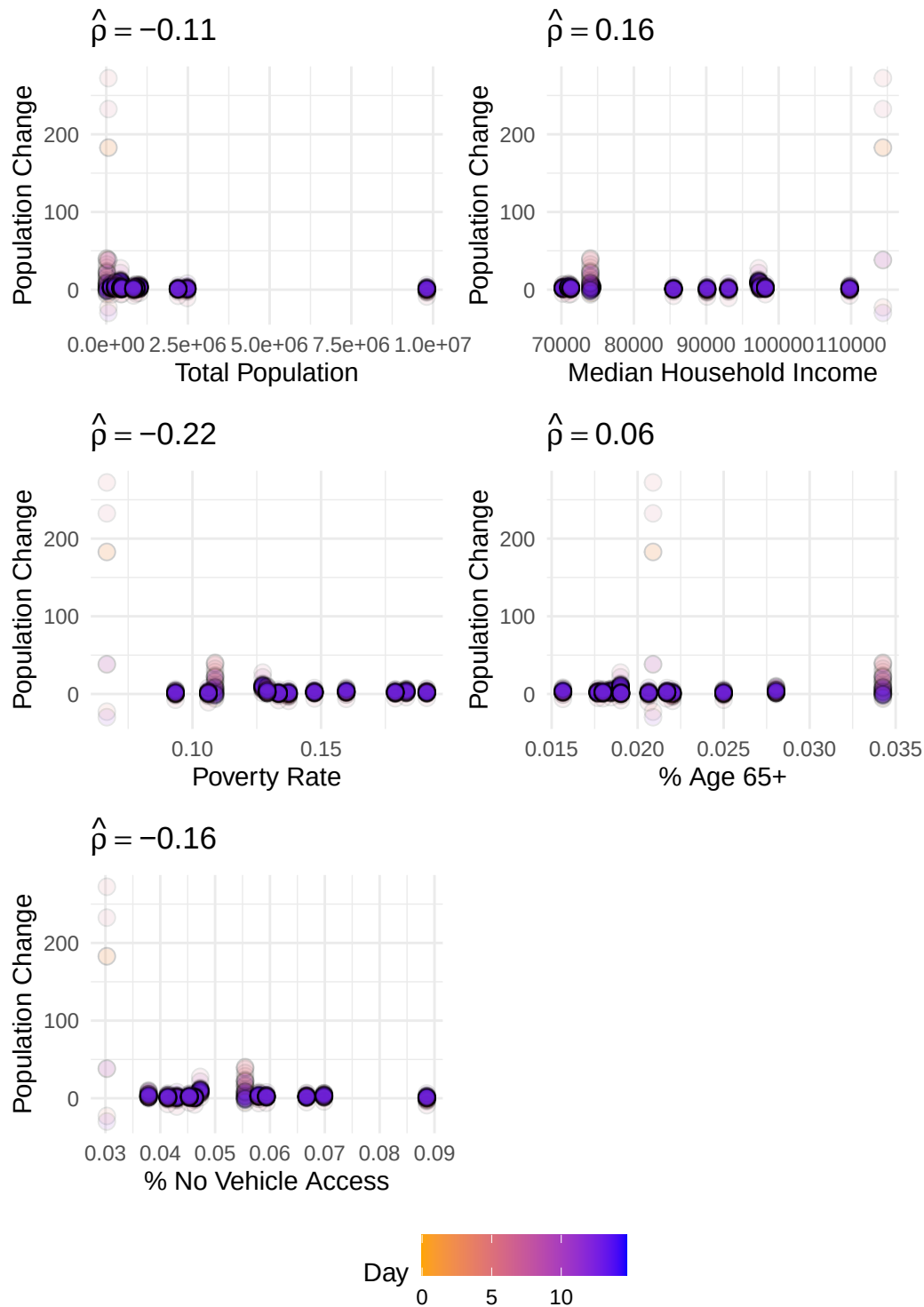


Figure 7: ... Population change is the percent change in Facebook Location Services-enabled user population counts per Bing tile in each county at 8-hour intervals. All demographic variables are already at the county level from the American Community Survey in 2022. Days are the 1st to 10th day of data collection, from May 19th to June 2nd.

Imputation Summary

All available dates and hours

Count	Original Count (Users)	Total (New) Count (Users)	Imputed Count (Users)	Share of Total (New) Count Imputed	Total Bing Tiles	Number of Imputed Bing Tiles	Share of Bing Tiles Imputed
Baseline	86,663,908	86,702,155	38,247	0.0%	251,966	540	0.2%
Crisis	87,408,197	87,502,715	94,518	0.1%	251,966	575	0.2%

Table 5: Share of baseline and crisis Facebook user counts and Bing tile observations that originate from imputed values for the selected reporting window. Counts represent the number of Facebook users with Location Services enabled who were observed within a Bing tile during a reporting window. Because the data are reported repeatedly across multiple 8-hour reporting windows, the same user may contribute to counts in multiple observations. To preserve observations suppressed by Meta’s privacy protections (i.e., Excluding counts for Bing tiles with counts less than 11), missing baseline counts were imputed as 3 and missing crisis counts were imputed as 9. The table reports both the share of total counts and the share of Bing tile observations that were affected by this imputation procedure.

References